
Tools, Not Talk: The Knowledge Ceiling of Biology Agents

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Abstract

1 Agentic systems for biology are increasingly built around orchestration: critics
2 that review answers, teams of role-playing scientists that debate, and models that
3 reason for longer. We show that on objectively scored biology research tasks this
4 machinery adds little beyond what matched independent sampling provides, and
5 we identify why. Across 150 pre-registered LAB-Bench and LABBench2 ques-
6 tions, compared at a matched budget of three model calls, self-critique never beats
7 simple majority voting, multi-agent debate helps only a model whose reasoning
8 is switched off, and longer reasoning does not improve accuracy even at $25\times$ the
9 generated tokens. The reason is a ceiling set by the questions themselves: which
10 question is asked explains 56% of the variance in correctness, while the choice of
11 model, strategy and reasoning effort explains 0.6%. That ceiling is largely missing
12 knowledge rather than missing deliberation. A single model call with web search
13 and code execution solves 47% of the questions that all eight systems in our main
14 study failed. For life-science agents, the budget belongs in tools, not talk.

15 1 Introduction

16 LLM agents now plan experiments, analyse data and generate hypotheses in chemistry and biology
17 [1, 2, 5, 6, 8, 16]. Many of these systems are organised as conversations: a critic reviews a draft,
18 several specialised scientists debate, an adjudicator decides. Their gains are usually reported against
19 a single model call, even though the agent makes many. Sampling several answers and voting already
20 improves accuracy [3, 12, 19], so these comparisons cannot tell whether the orchestration helps or
21 whether the extra calls do [9].

22 We separate the two. Every orchestrated strategy receives exactly the budget of a voting baseline:
23 three calls, the same model, the same prompts and the same decoding. We then ask what actually
24 moves accuracy on biology research tasks: orchestration, longer reasoning, or access to knowledge
25 through tools.

26 Contributions.

- 27 • **Orchestration adds little at matched compute.** Across three models, self-critique never
28 beats voting and breaks at least as many answers as it repairs. Debate helps only when the
29 model cannot reason on its own. Raising reasoning effort rarely helps either (§3.1–3.2).
- 30 • **The question decides.** Systems largely solve the same questions and fail the same ques-
31 tions. Question identity explains 56% of the variance in correctness and the system 0.6%
32 (§3.3).
- 33 • **Tools break the ceiling.** One tool-using call solves 47% of the questions that no main-
34 study system solved, and more compute on top of tools adds comparatively little (§3.4).

Table 1: Accuracy (%) at a matched budget of three calls (DIRECT uses one). Δ = DEBATE – VOTE in points, with paired bootstrap 95% CI.

Model	n	DIRECT	VOTE	CRITIQUE	DEBATE	Δ [95% CI]
GPT-OSS-120B	150	46.7	49.3	42.7	47.3	−2.0 [−8.7, +4.7]
Gemma 4 31B (no thinking)	150	48.7	48.7	47.3	57.3	+8.7 [+2.0, +15.3]
Flash-Lite	75	61.3	62.7	56.0	57.3	−5.3 [−13.3, +1.3]

2 Setup

Tasks. We draw 30 questions from each of five categories (150 in total): four from LAB-Bench [10] (protocol troubleshooting, sequence manipulation, database and literature questions; multiple choice) and open-ended sequence computations from LABBench2 SeqQA2 [11], scored with the official validators. All answers are scored deterministically, with no LLM judge.

Strategies. DIRECT makes one call. VOTE makes three independent calls and takes the majority answer. CRITIQUE solves, critiques its own solution, then revises. DEBATE has a molecular biologist and a skeptical reviewer solve independently, then an adjudicator decides. VOTE, CRITIQUE and DEBATE each use exactly three calls, and only visible outputs pass between calls (prompts in App. A).

Models. The matched-budget study uses GPT-OSS-120B [13] (low reasoning effort), Gemma 4 31B [17] (thinking disabled) and Gemini 3.5 Flash-Lite (low thinking) (GPT-OSS and Gemma on all 150 questions, Flash-Lite on a fixed 75). To test reasoning effort and tools, we add GPT-6 Luna, GPT-6 Sol and GPT-5.6 Terra, and rerun GPT-OSS and Flash-Lite at high effort, on a fixed 30-question subset with 6 per category. Task IDs, prompts, decoding and the primary analysis were frozen before evaluation (App. B). All contrasts are paired over questions, with bootstrap 95% CIs and exact McNemar tests.

3 Results

3.1 Orchestration adds little at matched compute

Three calls spent on orchestration generally do no better than three calls spent on independent samples (Table 1). **Self-critique does not beat voting** for any of the three models (−6.7, −1.3, −6.7 points; pooled −4.5 [−9.3, +0.3]). It breaks correct answers as readily as it fixes wrong ones: relative to the vote it rescues 24%, 13% and 25% of errors but corrupts 38%, 16% and 26% of correct answers. The damage is worst on exact sequence manipulation (SeqQA: −23 and −20 points for GPT-OSS and Flash-Lite), where a fluent but wrong critique argues the model out of a correct computation. **Debate helps only the model that cannot think.** For Gemma, whose thinking is disabled, debate beats voting by +8.7 points ($p = 0.015$); for the two models that reason, it does not (−2.0 and −5.3).

3.2 Longer reasoning does not substitute for knowledge

For most models, raising reasoning effort multiplies the cost without moving accuracy (Fig. 1). GPT-OSS at high effort generates $25\times$ more tokens and scores 47% against 50%; Flash-Lite spends $5\times$ more tokens and scores 67% against 70%. The one model that gains is GPT-6 Luna, which uses no reasoning tokens at low effort, and its high-effort single call reaches exactly the accuracy its low-effort debate already had (60%). Debate and reasoning thus act as substitutes, and neither gets past the plateau. Stacking orchestration on high effort adds little: GPT-6 Sol scores 70% with every strategy while the three-call strategies generate $\sim 3\times$ the tokens.

3.3 The question, not the system, decides

A two-way decomposition of the correctness matrix attributes 56% of the variance to the question and 0.6% to the system (Fig. 2). The pattern holds across all 33 model, strategy and effort configurations on the shared subset, frontier models at high effort included (47% vs. 3.9%). Systems largely

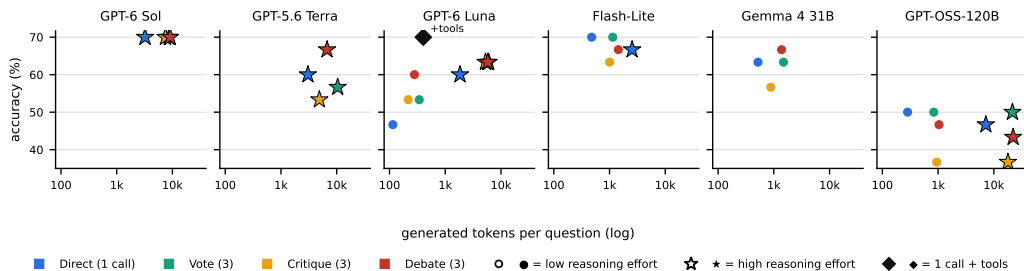


Figure 1: Every way of spending inference lands on a similar per-model plateau; tools rise above it. Accuracy vs. generated tokens per question on a shared 30-question set, one panel per model. Colour marks the strategy. Marker shape marks reasoning effort: circles (●) are low effort and stars (★) high effort. GPT-6 Sol and GPT-5.6 Terra were run at high effort only. Across nearly two orders of magnitude in tokens, accuracy moves little within a model. The black diamond (◆) is one GPT-6 Luna call with web search and code execution. It is the only configuration clearly above its model’s plateau, and among the cheapest.

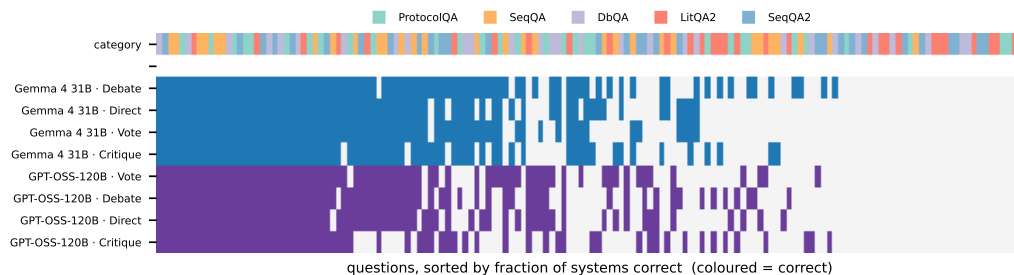


Figure 2: The question, far more than the system, decides. Correctness of every model × strategy system (rows) on every question (columns, sorted by the fraction of systems correct). Systems largely solve and fail the same questions. Question identity explains 56% [49, 62] of the variance in correctness; the system explains 0.6% [0.3, 1.9].

75 solve the same easy questions and fail the same hard ones. The 32 questions that no main-study
76 system solves are dominated by literature and database recall (18) and open-ended sequence com-
77 putation (9). These are failures of knowledge, which additional deliberation did not supply in any
78 configuration we tested.

79 3.4 Tools break the ceiling

80 Giving GPT-6 Luna web search and code execution in a single call raises accuracy on the shared
81 subset from 47% to 70% (+23 points [+7, +43], $p = 0.039$). That is more than any orchestration
82 or reasoning setting achieves, at ~ 404 generated tokens per question. On the 32 questions that all
83 eight main-study systems (two models × four strategies) failed, the same single call reaches 47%,
84 against 6% without tools (+41 points [+25, +56], $p = 0.0002$). Literature questions rise from 18%
85 to 91% and database questions from 0% to 43%. Retrieval helping on these categories is expected,
86 since they were designed for retrieval agents [10, 14]. The point is the contrast: none of the eight
87 main-study systems, whether single-call, voting or orchestrated, solved any of them. Compute on
88 top of tools adds comparatively little. High effort reaches 83% (+13, $p = 0.22$) and a three-call vote
89 77% (+7, $p = 0.73$), at $\sim 6\times$ and $\sim 3.5\times$ the tokens. No search or command reached the benchmark
90 or its datasets.

91 4 Implications for building life-science agents

92 **Benchmark against a matched vote.** A multi-call agent should beat independent sampling with the
93 same number of calls, not a single call [9]. Most orchestration gains disappear under this baseline,
94 consistent with compute-matched studies of debate in general reasoning [4, 15, 18, 20] and with the

95 unreliability of intrinsic self-correction [7]. **Spend the budget on tools, not talk.** Critics, debaters
 96 and longer chains of thought re-deliberate over knowledge the model does not have. Retrieval sup-
 97 plies it, at a fraction of the tokens. **Build domain tools for computation.** A generic shell barely
 98 helps open-ended sequence design and analysis (0% to 11%). The next gains lie in biology-specific
 99 tool ecosystems and specialist models, not in more conversation between copies of the same model.

100 **Scope.** Our tasks have deterministic answers, and we study short-horizon question answering
 101 rather than long-running discovery loops. The frontier and tool comparisons use a 30-question
 102 subset, and the tool condition adds a one-sentence note announcing the tools.

103 5 Conclusion

104 On biology research tasks, how a model spends extra inference barely changes which questions it
 105 gets right. This holds for voting, critique, debate and longer reasoning alike: the question explains
 106 56% of the variance in correctness and the system 0.6%. The ceiling is largely knowledge, and
 107 a single tool-using call lifts it where orchestration did not. Life-science agents should spend their
 108 budget on tools, not talk.

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176 A Prompts

177 All prompts are single user messages. {QUESTION} is the question (with protocol, options or injected files) and {ANSWER INSTRUCTION} is the format line for multiple choice or the official SeqQA2 suffix.

180 Direct / Independent×3 / Critique call 1 (solve)

181 Solve the following biology research question.
 182
 183 {QUESTION}
 184
 185 {ANSWER INSTRUCTION}
 186
 187 Respond in exactly this format and keep it short:
 188 <answer>YOUR ANSWER</answer>
 189 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
 190 JUSTIFICATION: <at most two sentences>

191 Critique call 2 (critique)

192 Below are a biology research question and a proposed solution. Act as a critical reviewer: identify any
 193 scientific, computational or reasoning error in the solution. If you think the answer is wrong, say which
 194 answer you believe is correct and why. If you find no error, say so.

195
196 {QUESTION}
197
198 Proposed solution:
199 {CALL 1 OUTPUT}
200
201 Respond with at most four sentences, starting with "CRITIQUE:".

202 **Critique call 3 (revise)**

203 Below are a biology research question, a first attempt at solving it, and a reviewer's critique of that
204 attempt. Either may contain mistakes. Considering both, produce the final answer.
205
206 {QUESTION}
207
208 First attempt:
209 {CALL 1 OUTPUT}
210
211 Reviewer's critique:
212 {CALL 2 OUTPUT}
213
214 {ANSWER INSTRUCTION}
215
216 Respond in exactly this format and keep it short:
217 <answer>YOUR ANSWER</answer>
218 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
219 JUSTIFICATION: <at most two sentences>

220 **Debate call 1 (researcher)**

221 You are a molecular biology researcher. Solve the following biology research question independently.
222
223 {QUESTION}
224
225 {ANSWER INSTRUCTION}
226
227 Respond in exactly this format and keep it short:
228 <answer>YOUR ANSWER</answer>
229 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
230 JUSTIFICATION: <at most two sentences>

231 **Debate call 2 (skeptic)**

232 You are a skeptical scientific reviewer. Solve the following biology research question independently, looking
233 especially for subtle experimental, biological or computational errors and traps.
234
235 {QUESTION}
236
237 {ANSWER INSTRUCTION}
238
239 Respond in exactly this format and keep it short:
240 <answer>YOUR ANSWER</answer>
241 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
242 JUSTIFICATION: <at most two sentences>

243 **Debate call 3 (adjudicator)**

244 Two scientists independently analysed the biology research question below. Adjudicate between their analyses
245 (either or both may be wrong) and submit the final answer.
246
247 {QUESTION}
248
249 Analysis 1 (molecular biology researcher):
250 {CALL 1 OUTPUT}
251
252 Analysis 2 (skeptical scientific reviewer):
253 {CALL 2 OUTPUT}

254 {ANSWER INSTRUCTION}
 255
 256
 257 Respond in exactly this format and keep it short:
 258 <answer>YOUR ANSWER</answer>
 259 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
 260 JUSTIFICATION: <at most two sentences>

261 B Pre-registration deviations

262 All task IDs, prompts, decoding settings and the primary analysis were frozen after a 10-question
 263 pilot and before any main-set call. Changes made after the freeze: (i) throttle and retry fixes for an
 264 undocumented 150 requests/hour provider limit (no completed call was re-run); (ii) 22 GPT-OSS
 265 questions served in full by a second provider of the same model (identical prompts, seeds and cap;
 266 excluding them leaves DEBATE – VOTE at -1.6), with partially served questions discarded and re-
 267 run on the primary provider; (iii) the reasoning-effort and tool arms on the fixed 30-question subset
 268 and the unsolved set, added after the freeze. Frontier models were served through their vendor CLI
 269 with web search, code execution and sub-agents disabled except in the tool condition (a first attempt
 270 with web search left on was discarded unanalysed); GPT-OSS at high effort was served by a second
 271 host whose low-effort accuracy matches the main run. The tool condition prepends one sentence
 272 announcing the tools, and all searches and commands were logged and checked for benchmark
 273 access (none).

274 C Additional results

Category	Gemma					Lite					OSS				
	D	I	C	M	Disagree	D	I	C	M	Disagree	D	I	C	M	Disagree
DbQA	56.7	53.3	40.0	56.7	33.3	40.0	40.0	40.0	33.3	26.7	46.7	46.7	43.3	36.7	36.7
LitQA2	33.3	30.0	33.3	36.7	46.7	60.0	53.3	53.3	46.7	26.7	20.0	30.0	40.0	36.7	43.3
ProtocolQA	56.7	56.7	60.0	70.0	20.0	66.7	80.0	86.7	73.3	26.7	56.7	50.0	43.3	46.7	33.3
SeqQA	56.7	56.7	66.7	73.3	46.7	73.3	73.3	53.3	66.7	13.3	63.3	66.7	43.3	60.0	33.3
SeqQA2	40.0	46.7	36.7	50.0	63.3	66.7	66.7	46.7	66.7	40.0	46.7	53.3	43.3	56.7	43.3

Table 2: Per-category accuracy (D=Direct, I=Independent×3, C=Critique, M=Multi). Numbers are percentages.